

Advanced Number Theory Introduction

①

$$3 = 52 \bmod 7 \Rightarrow 57 \div 7 = 7r3 \quad (7 \times 7 = 49)$$

* Euclidean theory $\Rightarrow \gcd(a, b)$

$$b - a \times m = r_1$$

$$\text{rem}(b \div a) = r_1$$

$$a - r_1 \times m_1 = r_2$$

$$\text{rem}(a \div r_1) = r_2$$

$$r_1 - r_2 \times m_2 = r_3$$

$$\text{rem}(r_1 \div r_2) = r_3$$

....

do until $r_{i+1} = 0$, then r_i is the gcd.

$$\begin{aligned} \gcd(16, 36) &\Rightarrow 36 - 16 \times 2 = 4 \\ 16 - 4 \times 4 &= 0 \end{aligned} \quad \therefore \text{gcd is } 4 //$$

* multiplication & addition are associative

$$\hookrightarrow 12 \equiv 3 \pmod{9}$$

$$22 \equiv 4 \pmod{9}$$

$$12 + 22 = 34$$

$$3 + 4 = 7$$

$$\left. \begin{array}{l} 12 + 22 = 34 \\ 3 + 4 = 7 \end{array} \right\} \begin{array}{l} 34 = 7 \pmod{9} \\ (34 = 27 + 7) \end{array}$$

$$12 \times 22 = 244$$

$$3 \times 4 = 12$$

$$\left. \begin{array}{l} 12 \times 22 = 244 \\ 3 \times 4 = 12 \end{array} \right\} \begin{array}{l} 244 = 12 \pmod{9} \\ \equiv 3 \pmod{9} \end{array}$$

$$(244 \neq 241 + 3)$$

* division in modular arithmetic is tricky

* exercise $a \equiv c \pmod{p}$, $a \div m \equiv c \div m \pmod{p}$

is this true?

consider $12 \equiv 3 \pmod{9}$, $22 \equiv 4 \pmod{9}$,

$$12 \times 22 \equiv 12 \pmod{9},$$

but it doesn't mean $22 \equiv 1 \pmod{9}$.

* Bezant's law/lemma / extended euclidean algorithm.

$$* \gcd(65, 30) = 5$$

$$\Rightarrow 65 - 30 \times 2 = 5$$

$$30 - 5 \times 6 = 0$$

$\therefore$ gcd is 5

if $\gcd(a, b) = c$,

then, $m \times a + n \times b = c$.

using the bezant's ~~algo~~ lemma $\Rightarrow$

$$5 = \underline{1} \times 65 + \underline{-2} \times 30$$

* example $1 \Rightarrow \gcd(182, 35) = 7$

$$182 - 35 \times 5 = 7$$

$$\therefore \gcd(182, 35) = 7$$

$$35 - 7 \times 5 = 0$$

bezant's coeffs $\Rightarrow$

$$7 = 182 \times \underline{1} + \underline{-5} \times 35$$